

PROBLEM STATEMENT #32 | Retail, E-Commerce and Finance

Personal Subscription and Recurring Expense Auditor

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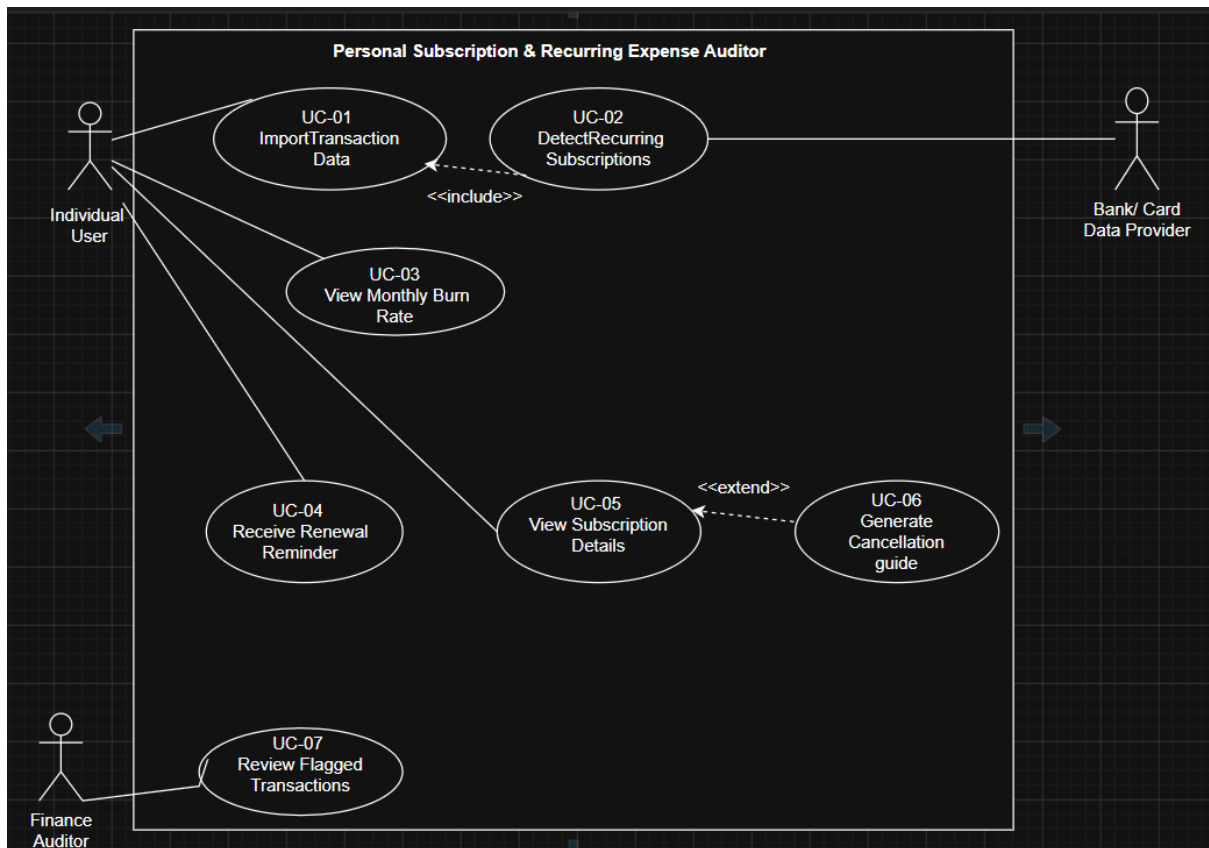
SECTION: G

1.Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall analyze transaction logs to identify recurring billing patterns and project total monthly subscription expenditure.	High	Pass: Recurring services are identified and the monthly subscription expenditure is calculated. Fail: A one-time payment is incorrectly identified as a recurring subscription.	This is the core functionality of the system.
FR-002	Functional	The system shall allow the user to import transaction data in CSV or OFX format.	High	Pass: A valid CSV/OFX file is successfully imported and its transactions are displayed. Fail: An invalid or unsupported file is accepted.	Transaction data is required for identifying recurring subscriptions.
FR-003	Functional	The system shall identify upcoming subscription renewal dates and alert the user before the renewal date.	High	Pass: The user receives an alert before the recorded renewal date. Fail: No alert is generated for an upcoming renewal.	Helps users avoid unwanted subscription renewals.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall calculate and display the user's monthly recurring subscription burn rate.	High	Pass: The displayed burn rate matches the total cost of active recurring subscriptions for the month. Fail: The displayed amount does not match the calculated recurring subscription total.	Helps users understand their monthly subscription spending.
FR-005	Functional	The system shall provide a cancellation guide for a selected recurring subscription.	Medium	Pass: The system displays cancellation steps when a user selects a subscription. Fail: No cancellation guidance is displayed.	Helps users cancel unwanted subscriptions easily.
NFR-001	Non-Functional — Performance & Security	The system shall parse CSV/OFX financial transaction imports securely without storing raw banking credentials in cleartext.	High	Pass: Testing confirms that valid transaction files are parsed and raw banking credentials are not stored in cleartext. Fail: Raw banking credentials are stored in cleartext.	Protects sensitive financial information.
NFR-002	Non-Functional — Performance	The system shall display transaction analysis results within 3 seconds for files containing up to 10,000 transactions.	High	Pass: At least 95% of test files containing up to 10,000 transactions produce results within 3 seconds. Fail: More than 5% of tests exceed 3 seconds.	Ensures the application remains responsive while processing transaction data.

2.UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Analyze Transactions

Use Case ID: UC-02

Use Case Name: Analyze Transactions

Primary Actor: Finance Auditor

Supporting Actor: Individual User

Preconditions

1. The user has access to the Personal Subscription & Recurring Expense Auditor system.
2. Transaction data has been imported into the system in CSV or OFX format.
3. The imported transaction data is available for analysis.

Postconditions

1. The transaction data has been successfully analyzed.
2. Recurring billing patterns have been identified.
3. Recurring subscriptions have been listed.
4. The total monthly subscription expenditure has been calculated and displayed.

Main Success Scenario

1. The Finance Auditor selects **Analyze Transactions**.
2. The system retrieves the imported transaction data.
3. The system validates the available transaction records.
4. The system analyzes transaction dates, amounts, and billing patterns.
5. The system identifies transactions that occur repeatedly.
6. The system classifies the identified recurring transactions as potential subscriptions.
7. The system calculates the total monthly subscription expenditure.
8. The system displays the identified recurring subscriptions and the calculated monthly expenditure.
9. The Finance Auditor reviews the analysis results.
10. The use case ends successfully.

Alternate Flow

4a. Invalid or Insufficient Transaction Data

1. The system detects that the transaction data is invalid, incomplete, or insufficient for analysis.
2. The system displays an appropriate error message.
3. The system asks the user to provide valid transaction data.
4. The user imports a valid CSV or OFX transaction file.
5. The system validates the new data.
6. The system resumes the analysis from Step 4 of the Main Success Scenario.

Result

The system provides the Finance Auditor with a list of identified recurring subscriptions and the projected total monthly subscription expenditure.